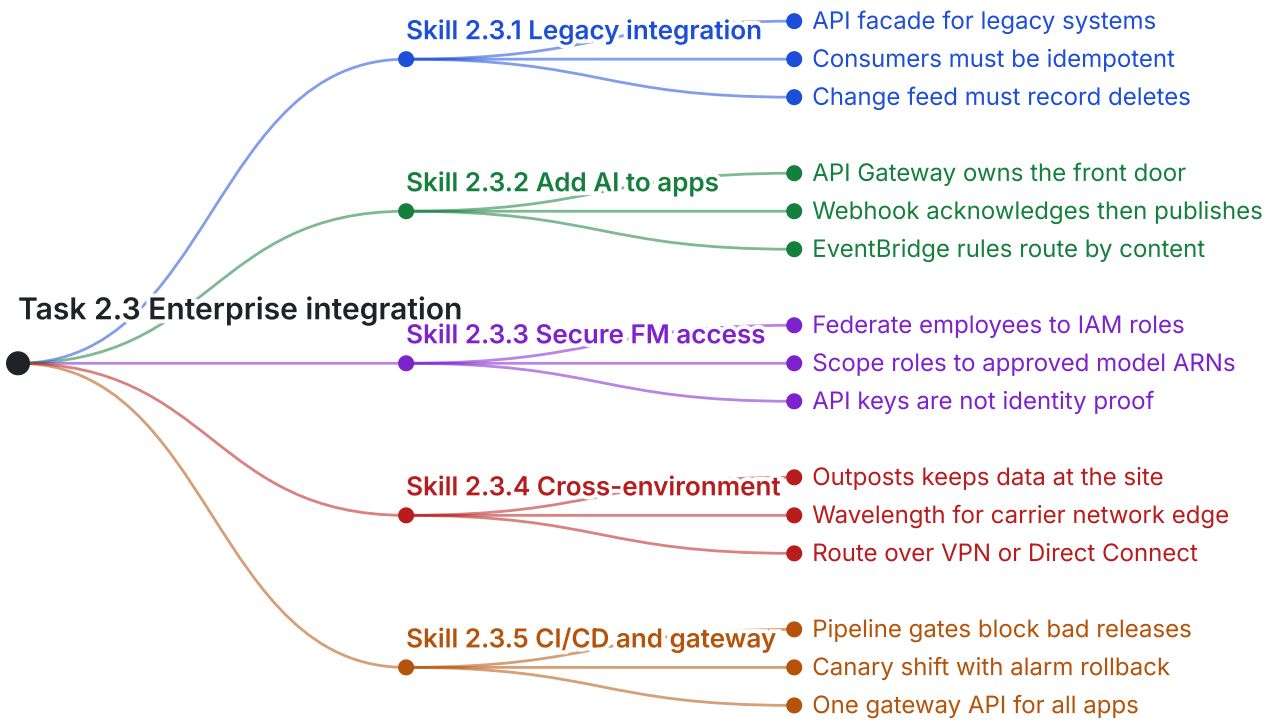
Memory sheet: Task 2.3 Enterprise integration architectures

Domain 2, weight 26 percent. Read this sheet, then answer the flash questions aloud.



Deciding rules

When the stem says	Think of	Not when
The caller waits for the model reply	Synchronous API path with REST or HTTP and Lambda	The producer needs no reply and work can finish later
The reply streams in pieces to a chat client	REST API with STREAM response transfer	The design uses an HTTP API, which has no response streaming
One event must reach many consumers that work later	EventBridge rules or an SNS topic fan-out	The caller needs the answer in the same request
A backlog must be visible and paced by workers	SQS queue with workers that scale on queue depth	One event starts one function and no backlog view is needed
An on-premises system needs steady high bandwidth	Direct Connect private virtual interface with VPN backup	A fast setup over an internet path is acceptable
Data must stay at a customer site and AWS APIs must work there	AWS Outposts with a local gateway	The users are mobile users on a carrier network
Employees sign in with corporate logins	Federation from the identity provider to IAM roles	Each developer would get a long-lived IAM user key
A release must be gated before production	CodePipeline stages with CodeBuild test actions	A checklist kept by developers or a deploy from a laptop

Etch lines

1. Acknowledge the partner first, then publish the event, and let consumers call the model.
2. Response streaming is offered on REST APIs only, never on HTTP APIs.
3. Event delivery is usually at least once, so consumers must be idempotent.
4. Federate employees to IAM roles and do not create IAM users for them.
5. Use an inbound resolver endpoint when the site must resolve AWS private names.
6. Rolling back code alone leaves the bad prompt live.

Flash questions

- Q: Can an HTTP API stream a model reply to a chat client? A: No, response streaming is offered on REST APIs only.
- Q: How should a webhook handler treat a model call that runs longer than the partner allows? A: Acknowledge the partner, publish an event, and let a consumer call the model.
- Q: What delivery guarantee should shape how consumers handle duplicate events? A: Usually at least once, so consumers must be idempotent.
- Q: Which service routes one event to different targets based on its content? A: EventBridge, with rules that match the event pattern and send it to targets.
- Q: Which Direct Connect virtual interface type reaches a VPC? A: A private virtual interface.
- Q: Why is an API key not enough to prove who called the API? A: A key identifies a client application for quota purposes, not a person.
- Q: What does CodePipeline model that CodeBuild alone does not? A: Ordered release stages and approval gates across the whole release.
- Q: What does a GenAI gateway do before it calls a model? A: It redacts personal data, loads the approved prompt, and applies a guardrail.

Source: Task 2.3 deciding constraint, Skills 2.3.1 to 2.3.5, Versus sections, and Read a long question. AWS docs and the study guide sections above, no practice-test text.